

Year 8 Science – Booklet 24: Chemistry

Reactivity Series — ANSWERS

Note: this booklet's content is identical to Year 8 Booklet 7 (Reactivity Series), so the same worked answers apply throughout.

Quick Test (p.65)

1. By electrolysis.
2. By observing how vigorously each metal reacts with air (oxygen), water, and dilute acid.
3. A metal oxide.
4. Zinc oxide.
5. $2\text{Zn} + \text{O}_2 \rightarrow 2\text{ZnO}$
6. Potassium hydroxide and hydrogen.
7. $2\text{K} + 2\text{H}_2\text{O} \rightarrow 2\text{KOH} + \text{H}_2$

Section A – Multiple Choice (5 marks)

1. c) magnesium oxide
2. a) magnesium
3. d) metal hydroxide and hydrogen
4. c) calcium hydroxide and hydrogen
5. a) hydrogen

Section B (20 marks)

1. Complete the word equations

- a) magnesium + oxygen → magnesium oxide
- b) iron + oxygen → iron oxide
- c) copper + oxygen → copper oxide
- d) potassium + water → potassium hydroxide + hydrogen
- e) sodium + water → sodium hydroxide + hydrogen
- f) calcium + water → calcium hydroxide + hydrogen
- g) magnesium + hydrochloric acid → magnesium chloride + hydrogen
- h) magnesium + nitric acid → magnesium nitrate + hydrogen
- i) calcium + hydrochloric acid → calcium chloride + hydrogen
- j) zinc + hydrochloric acid → zinc chloride + hydrogen

2. Complete the symbol equations

- a) $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$
- b) $2\text{Fe} + \text{O}_2 \rightarrow 2\text{FeO}$
- c) $\text{C} + \text{O}_2 \rightarrow \text{CO}_2$

Section C (8 marks)

1. Teacher's table (silver, sodium, magnesium, zinc)

- a). Most reactive → least reactive: **Sodium, Magnesium, Zinc, Silver.**
- b). Hydrogen.

c). Magnesium chloride.

d) i). A metal not in the table that reacts like silver (no reaction): gold (or another very unreactive metal, e.g. platinum).

d) ii). A metal not in the table that reacts like sodium (vigorous with water): potassium (or calcium/lithium).

2. *Calcium + hydrochloric acid* → *calcium chloride + hydrogen*

a). Calcium (or hydrogen) – the two elements in the equation; hydrochloric acid and calcium chloride are compounds.

b). A metal that reacts more vigorously than calcium: sodium (or potassium).

c). A metal that reacts less vigorously than calcium: magnesium (or zinc, iron, lead, copper).